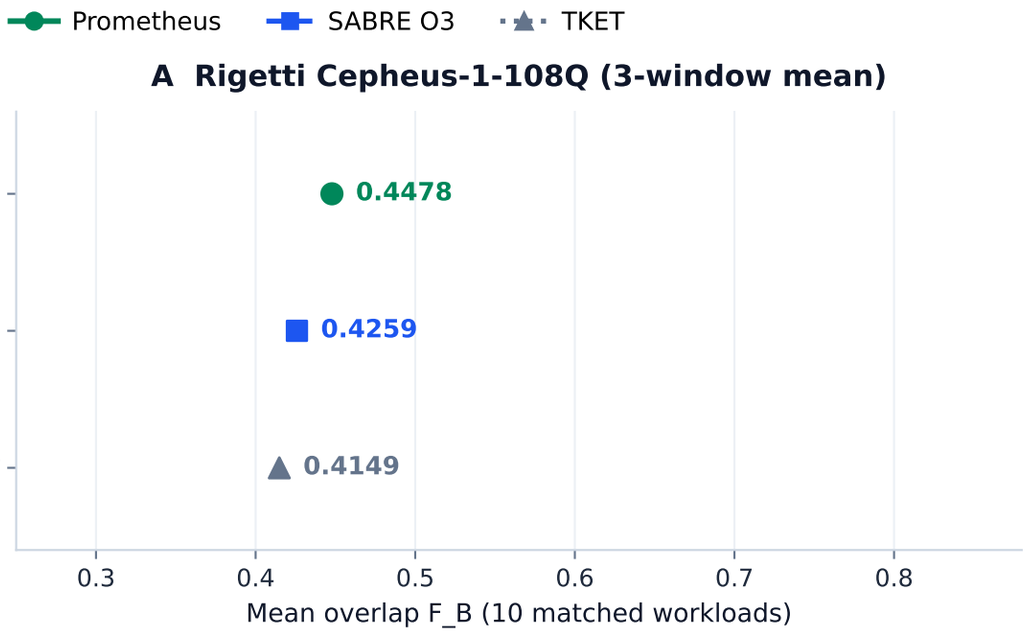
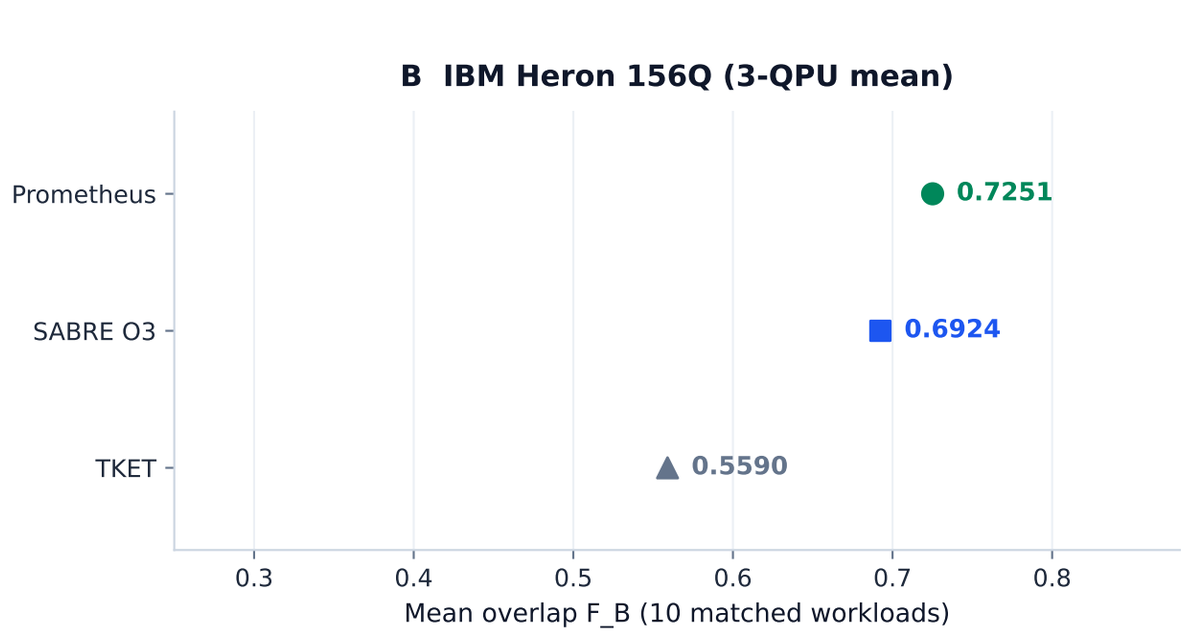
Prometheus: cross-provider hardware validation

10 identical logical workloads (6-12 qubits) | Rigetti Cepheus-1-108Q (3 windows) vs IBM Heron (3 QPUs)

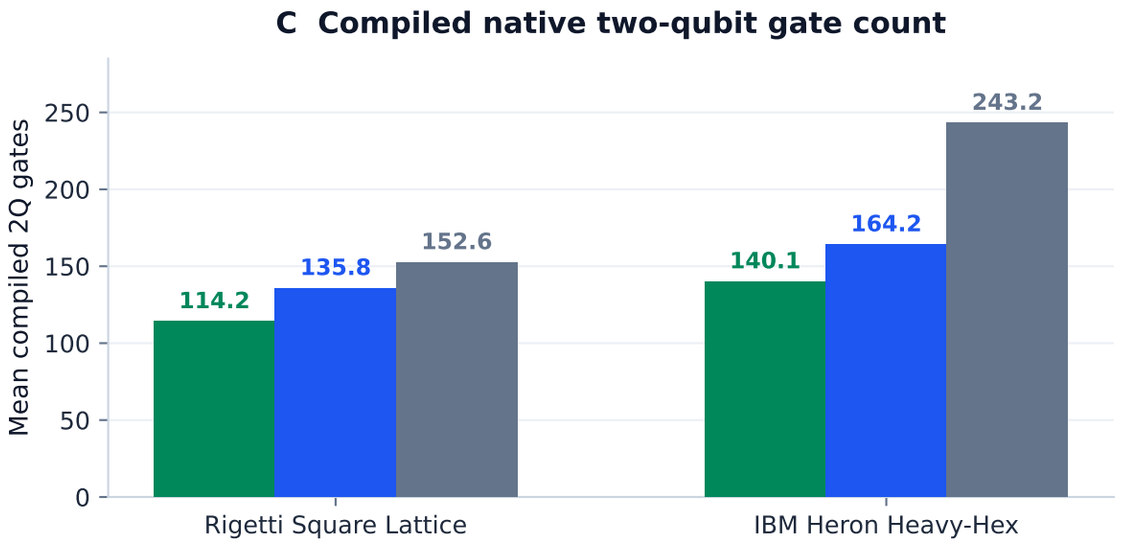
Comparing 4-tile planar square lattice (Rigetti, native CZ) and heavy-hex lattice (IBM Heron, 156 qubits).



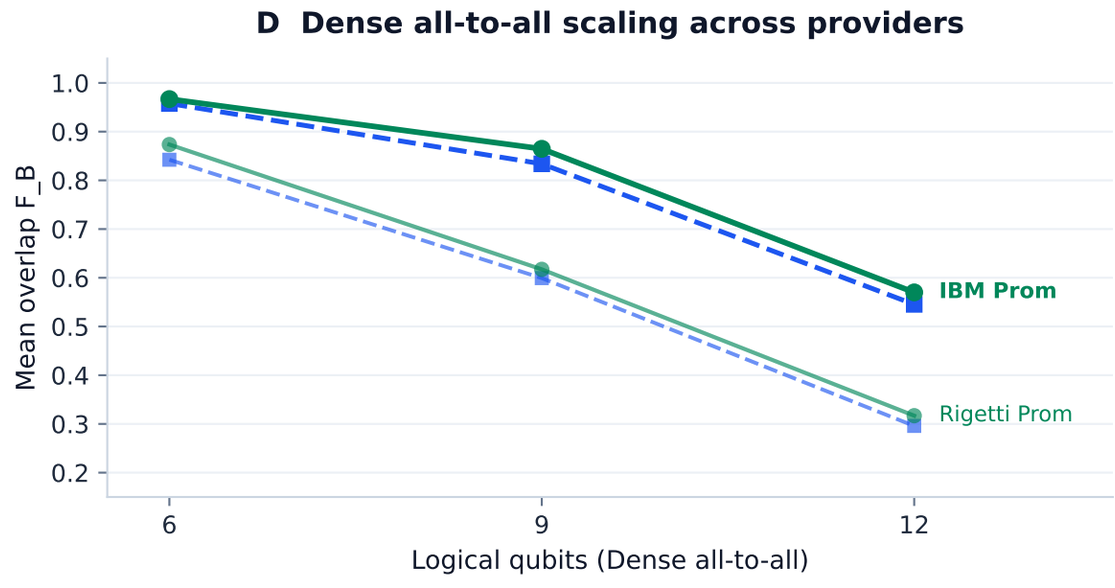
Prometheus gains +5.15% vs SABRE and +7.95% vs TKET.



Prometheus gains +4.73% vs SABRE and +29.71% vs TKET.



Total compiled 2Q gates across 10 workloads (114.2 on Rigetti vs 140.1 on IBM).



Upper pair: IBM Heron (3 QPUs); lower pair: Rigetti Cepheus-1-108Q (3 windows).

Cross-provider overlap gain: +5.15% vs SABRE on Rigetti | +4.73% vs SABRE on IBM Heron

Vs TKET: +7.95% on Rigetti (0.4478 vs 0.4149) | +29.71% on IBM Heron (0.7251 vs 0.5590). N = 10 matched workloads.

Prometheus reduces compiled two-qubit gates on both topologies: -15.91% vs SABRE (-25.16% vs TKET) on Rigetti; -14.65% vs SABRE (-42.38% vs TKET) on IBM.